

**Reference manual:
Communication
protocol
specifications**

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Reference Manual, Communication Protocol Specifications

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List of other communication protocols

For further information about protocol specifications, refer to the following code numbers:

Protocol	Code number
ASTM6xx	990-487
ASTM	990-490
HL7 2.5	902-896
HL7 HISLIS PLUS	990-492
HL7	990-493
Low level Protocols	990-544
NPT Protocols	994-039
TCM Protocols	994-038
Reference Manual	994-188

Analyzer and protocol overview

Introduction The following gives an overview of which protocol is used by which analyzer. The chart also shows which protocol the RADIANCE System can use to connect with the LIS. Given below are details on **Analyzer specific protocols**.

ANALYZER	CONNECTION	PROTOCOL							
		HIS LIS PLUS HL7	HL 7 v2. 2	HL7 v2. 5	HIS LIS PLUS	AST M 6xx	ASTM	POC T 1-A DML	Propriety
ABL9	AQURE								x
	LIS		x	x			x		
ABL90 FLEX	RADIANCE								x
	LIS		x	x		x	x	x	
ABL800 FLEX series	RADIANCE								x
	LIS		x	x		x	x	x	
ABL80 FLEX	RADIANCE								x
	LIS		x	x		x	x	x	
ABL800 Basic	RADIANCE								x
	LIS		x	x		x	x	x	
ABL700 Series	RADIANCE								x
	LIS		x			x	x		
ABL77	RADIANCE								x
	LIS		x				x		
RADIANCE	LIS	x	x	x	x	x	x		
AQT 90 FLEX	RADIANCE								x
	LIS		x	x		x	x	x	
POC analyzer	RADIANCE							x	

**Analyzer
specific
protocols**

The following have their own analyzer specific protocol:

ABL5xx/6xx analyzer

NPT7 analyzer

TCM monitor

List of parameters and units

Introduction The following contains two lists of parameters and units, one for the ABL analyzers (ABL700, ABL800, ABL80 FLEX, ABL90 FLEX, ABL9) and one for the AQT90 FLEX analyzer.

List of parameters and units for the ABL FLEX analyzers

Patient results The following gives a list of parameters and unit names for patient results for the ABL800 analyzer.

Parameter	ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
Baro.	B	B	mmHg	kPa	torr [†]
pH	pH	pH			
pH(T)	pH(T)	pH(T)			
pCO ₂	pCO2	pCO2	mmHg	kPa	torr [†]
pCO ₂ (T)	pCO2(T)	pCO2(T)	mmHg	kPa	torr [†]
cHCO ₃ ⁻ (P)	HCO3-	HCO3-	mmol/L		
cBase(B)	ABE	ABE	mmol/L		
cBase(Ecf)	SBE	SBE	mmol/L		
cHCO ₃ ⁻ (P,st)	SBC	SBC	mmol/L		
ctCO ₂ (P)	tCO2(P)	tCO2(P)	mL/dL	mmol/L	Vol%
ctCO ₂ (B)	tCO2(B)	tCO2(B)	mL/dL	mmol/L	Vol%
pH(st)	pH(st)	pH(st)			
VO ₂ / V (Dry Air) [†]	O2	O2	%	fraction	
VCO ₂ / V (Dry Air)	CO2	CO2	%	fraction	
ctHb	tHb	tHb	g/dL	g/L	mmol/L
FHHb [‡]	RHb	RHb	%	fraction	
FO ₂ Hb [‡]	O2Hb	O2Hb	%	fraction	
sO ₂	sO2	sO2	%	fraction	
FCOHb [‡]	COHb	COHb	%	fraction	
FMetHb [‡]	MetHb	MetHb	%	fraction	
FHbF [†]	HbF	HbF	%	fraction	

Parameter	ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
Hct	Hct	Hct	%	fraction	
pO_2	pO2	pO2	mmHg	kPa	torr [†]
$pO_2(T)$	pO2(T)	pO2(T)	mmHg	kPa	torr [†]
$pO_2(A)$	pO2(A)	pO2(A)	mmHg	kPa	torr [†]
$pO_2(A, T)$	pO2(A),T	pO2(A),T	mmHg	kPa	torr [†]
$p50^{\ddagger}$	p50(act)	p50(act)	mmHg	kPa	torr [†]
$p50(T)^{\ddagger}$	p50(act),T	p50(act),T	mmHg	kPa	torr [†]
$p50(st)^{\ddagger}$	p50(st)	p50(st)	mmHg	kPa	torr [†]
$pO_2(A-a)$	AaDpO2	AaDpO2	mmHg	kPa	torr [†]
$pO_2(A-a, T)$	AaDpO2,T	AaDpO2,T	mmHg	kPa	torr [†]
$pO_2(a/A)$	a/ApO2	a/ApO2	%	fraction	
$pO_2(a/A, T)$	a/ApO2,T	a/ApO2,T	%	fraction	
$pO_2(x)^{\ddagger}$	px	px	mmHg	kPa	torr [†]
$pO_2(S^0)^{\ddagger}$	pO2(S)	pO2(S0)	mmHg	kPa	torr
ctO ₂ (B)	tO2	tO2	mL/dL	mmol/L	Vol%
ctO ₂ (a- $\bar{v}$) [†]	avDO2	ctO2(a-v)	mL/dL	mmol/L	Vol%
BO ₂ [‡]	O2CAP	O2CAP	mL/dL	mmol/L	Vol%
ctO ₂ (x) [†]	cx	cx	mL/dL	mmol/L	Vol%
$\dot{D} O_2^{\ddagger}$	DO2	DO2	mL/min	mmol/min	
$\dot{Q}_t^{\ddagger}$	Qt	Qt	L/min		
$\dot{V}O_2^{\ddagger}$	VO2	VO2	mL/min	mmol/min	
FO ₂ (I)	FIO2	FIO2	%	fraction	
FShunt [‡]	Shunt	Shunt	%	fraction	
FShunt(T) [‡]	Shunt,T	Shunt,T	%	fraction	
RI	RI	RI	%	fraction	
RI(T)	RI,T	RI,T	%	fraction	
RQ [‡]	RQ	RQ			
Qx [‡]	Qx	Qx			

Parameter	ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
$sO_2(P)^{\dagger}$	sO2(P)	sO2(P0)	%	fraction	
$VCO^{\ddagger}$	V(CO)	VCO	mL		
$V(B)^{\dagger}$	V(B)	V(B)	L		
cK^{+}	K+	K+	mmol/L	meq/L	
cNa^{+}	Na+	Na+	mmol/L	meq/L	
cCa^{2+}	Ca++	Ca++	mmol/L	meq/L	mg/dL
$cCa^{2+}(7.4)$	Ca(7.4)	Ca(7.4)	mmol/L	meq/L	mg/dL
cCl^{-}	Cl-	Cl-	mmol/L	meq/L	
AnionGap, K^{+}	Anion gap (K+)	Anion gap (K+)	mmol/L	meq/L	
Anion Gap	Anion gap	Anion gap	mmol/L	meq/L	
Temp	T	T	Cel	F	
$pO^{\dagger}$	P	P0	mmHg	kPa	torr
$S^{0\dagger}$	S	S0	%	fraction	
$pO_2(\bar{V})^{\dagger}$	pO2(v)	pO2(v)	mmHg	kPa	torr [†]
$sO_2(\bar{V})^{\dagger}$	sO2(v)	sO2(v)	%	fraction	
$cGlu^{\ddagger}$	Glu	Glu	mmol/L	mg/dL	
cLac	Lac	Lac	mmol/L	meq/L	mg/dL
$cK^{+}(U)^{\dagger}$	K+(urine)	cK+(U)	mmol/L	meq/L	
$cNa^{+}(U)^{\dagger}$	Na+(urine)	cNa+(U)	mmol/L	meq/L	
$cCl^{-}(U)^{\dagger}$	Cl-(urine)	cCl-(U)	mmol/L	meq/L	
$pO_2(x,T)^{\ddagger}$	px,T	px,T	mmHg	kPa	torr [†]
$FCOHb(1)^{\ddagger}$	COHb(1)	FCOHb(1)	%	fraction	
$FCOHb(2)^{\ddagger}$	COHb(2)	FCOHb(2)	%	fraction	
ctBil [†]	tBil	tBil	micromol/L	mg/dL	mg/L
mOsm [‡]	mOsm	mOsm	mmol/kg		
cH^{+*}	cH+	cH+	nmol/L		
ABE	ABE	ABE	mmol/L		
SBE	SBE	SBE	mmol/L		
$cH^{+}(T)^{*}$	cH+(T)	cH+(T)	nmol/L		

Parameter	ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
$pO_2(a)/FO_2(l)$	pO2(a)/FIO2	pO2(a)/FIO2	mmHg	kPa	torr [†]
$pO_2(a,T)/FO_2(l)$	pO2(a,T)/FIO2	pO2(a,T)/FIO2	mmHg	kPa	torr [†]
cBase(B,ox)	cBase(B,ox)	cBase(B,ox)	mmol/L		
cBase(Ecf,ox)	cBase(Ecf,ox)	cBase(Ecf,ox)	mmol/L		
cCrea [†]	Crea	Crea	micromol/L	mg/dL	
cCreatine [†]	Creatine	Creatine	micromol/L	mg/dL	
GFR if nonAA [†]	GFR_nonAA	GFR_nonAA	mL/min/1.73m ²		
GFR if AA [†]	GFR_AA	GFR_AA	mL/min/1.73m ²		

* Parameters and units are not available on the ABL80 FLEX analyzer

[†] Parameters and units are not available on the ABL80 FLEX or ABL9 analyzers

[‡] Parameters and units are not available on the ABL9 analyzer

QC results The following gives a list of parameter and unit names for QC results.

Parameter	ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
Baro. [†]	B	B	mmHg	kPa	torr
pH	pH(T)	pH(T)			
pCO ₂	pCO2(T)	pCO2(T)	mmHg	kPa	torr [†]
pO ₂	pO2(T)	pO2(T)	mmHg	kPa	torr [†]
ctHb [‡]	tHb	tHb	g/dL	g/L	mmol/L
FO ₂ Hb [‡]	O2Hb	O2Hb	%	fraction	
sO ₂ [‡]	sO2	sO2	%	fraction	
FCOHb [‡]	COHb	COHb	%	fraction	
FMetHb [‡]	MetHb	MetHb	%	fraction	
FHbF [‡]	HbF	HbF	%	fraction	
cK ⁺	K+	K+	mmol/L	meq/L	
cNa ⁺	Na+	Na+	mmol/L	meq/L	
cCa ²⁺	Ca++	Ca++	mmol/L	meq/L	mg/dL
cCl ⁻	Cl-	Cl-	mmol/L	meq/L	
Temp	T	T	Cel	F	
cGlu [‡]	Glu	Glu	mmol/L	mg/dL	
cLac	Lac	Lac	mmol/L	meq/L	mg/dL
ctBil [†]	tBil	tBil	micromol/L	mg/dL	mg/L
cCrea [†]	Crea(T)	Crea(T)	micromol/L	mg/dL	
cCreatine [†]	Creatine(T)	Creatine(T)	micromol/L	mg/dL	
cCrea diff [†]	Crea(diff)	Crea(diff)	micromol/L	mg/dL	
Hct	Hct	Hct	%	Fraction	

[†] Parameters and units are not available on the ABL80 FLEX or ABL9 analyzers

[‡] Parameters and units are not available on the ABL9 analyzer

Calibration results The following gives the parameter and unit names for calibration results.

ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
pH ^ 1	pH ^ High			
pH ^ 2	pH ^ Low			
pH ^ Drift1	pH ^ DriftH			
pH ^ Drift2	pH ^ DriftL			
pH ^ Status	pH ^ Status			
pH ^ Sens [†]	pH ^ Sens	%	fraction	
pCO2 ^ 1	pCO2 ^ Low	mmHg	kPa	torr
pCO2 ^ 2	pCO2 ^ High	mmHg	kPa	torr
pCO2 ^ Drift1	pCO2 ^ DriftL	mmHg	kPa	torr
pCO2 ^ Drift2	pCO2 ^ DriftH	mmHg	kPa	torr
pCO2 ^ Status	pCO2 ^ Status	mmHg	kPa	torr
pCO2 ^ Sens [†]	pCO2 ^ Sens	%	fraction	
pO2 ^ 1	pO2 ^ High	mmHg	kPa	torr
pO2 ^ 2	pO2 ^ Low	mmHg	kPa	torr
pO2 ^ Drift1	pO2 ^ DriftH	mmHg	kPa	torr
pO2 ^ Drift2	pO2 ^ DriftL	mmHg	kPa	torr
pO2 ^ Sens [†]	pO2 ^ Sens	pA/mmHg	pA/kPa	
pO2 ^ Zero	pO2 ^ Zero	mmHg	kPa	torr
tHb ^ Zero [‡]	tHb ^ Zero	pA	nA	
tHb ^ ZeroDrift [‡]	tHb ^ ZeroDrift	pA	kPa	torr
tHb ^ ZeroStatus [‡]	tHb ^ ZeroStatus			
K+ ^ 1	K+ ^ Low	mmol/L	meq/L	mg/L
K+ ^ 2	K+ ^ High	mmol/L	meq/L	torr
K+ ^ Drift1	K+ ^ DriftL	mmol/L	meq/L	torr
K+ ^ Drift2	K+ ^ DriftH	mmol/L	meq/L	torr
K+ ^ Status	K+ ^ Status	mmol/L	meq/L	torr
K+ ^ Sens [†]	K+ ^ Sens	%	fraction	

ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
Na+ ^ 1	Na+ ^ High	mmol/L	meq/L	torr
Na+ ^ 2	Na+ ^ Low	mmol/L	meq/L	torr
Na+ ^ Drift1	Na+ ^ DriftH	mmol/L	meq/L	torr
Na+ ^ Drift2	Na+ ^ DriftL	mmol/L	meq/L	torr
Na+ ^ Status	Na+ ^ Status	mmol/L	meq/L	torr
Na+ ^ Sens [†]	Na+ ^ Sens	%	fraction	
Ca++ ^ 1	Ca++ ^ Low	mmol/L	meq/L	mg/dL
Ca++ ^ 2	Ca++ ^ High	mmol/L	meq/L	mg/dL
Ca++ ^ Drift1	Ca++ ^ DriftL	mmol/L	meq/L	mg/dL
Ca++ ^ Drift2	Ca++ ^ DriftH	mmol/L	meq/L	mg/dL
Ca++ ^ Status	Ca++ ^ Status	mmol/L	meq/L	mg/dL
Ca++ ^ Sens [†]	Ca++ ^ Sens	%	fraction	
Cl- ^ 1	Cl- ^ High	mmol/L	meq/L	
Cl- ^ 2	Cl- ^ Low	mmol/L	meq/L	
Cl- ^ Drift1	Cl- ^ DriftH	mmol/L	meq/L	
Cl- ^ Drift2	Cl- ^ DriftL	mmol/L	meq/L	
Cl- ^ Status	Cl- ^ Status	mmol/L	meq/L	
Cl- ^ Sens [†]	Cl- ^ Sens	%	fraction	
Glu ^ Sens [#]	Glu ^ Sens	pA/mM		
Glu ^ Drift [‡]	Glu ^ Drift	mmol/L	mg/dL	
Lac ^ Sens	Lac ^ Sens	pA/mM		
Lac ^ Drift	Lac ^ Drift	mmol/L	meq/L	mg/dL
Glu ^ 1 [‡]	Glu	mmol/L	mg/dL	
Lac ^ 1	Lac	mmol/L	meq/L	mg/dL
Crea ^ 1 [§]	Crea ^ 1	micromol/L	mg/dL	
Crea ^ 2 [§]	Crea ^ 2	micromol/L	mg/dL	
Crea ^ 3 [§]	Crea ^ 3	micromol/L	mg/dL	
Crea ^ Drift1 [§]	Crea ^ Drift1	pA/uM	pA/(mg/dL)	
Crea ^ Drift2 [§]	Crea ^ Drift2	%	fraction	
Crea ^ Drift3 [§]	Crea ^ Drift3	%	fraction	

ASTM / HL7	ASTM6xx	Unit 1	Unit 2	Unit 3
Crea ^ Sens1 [§]	Crea ^ Sens1	pA/uM	pA/(mg/dL)	
Crea ^ Sens2 [§]	Crea ^ Sens2	%	fraction	
Crea ^ Sens3 [§]	Crea ^ Sens3	%	fraction	
Creatine ^ 2 [§]	Creatine ^ 2	micromol/L	mg/dL	
Creatine ^ Drift2 [§]	Creatine ^ Drift2	%	fraction	
Creatine ^ Sens2 [§]	Creatine ^ Sens2	%	fraction	
Glu ^ Sens80 [‡]	Glu ^ Sens80			
OXI _ WatCurrent [*]	tHb ^ Status	pA		
Glu ^ Status [*]	Glu ^ Status	pA		
Lac ^ Status [*]	Lac ^ Status	pA		
tHb ^ Sens [*]	tHb ^ Sens	%	fraction	
Glu ^ 0 [*]	Glu ^ Low	mmol/L	mg/dL	
Lac ^ 0 [*]	Lac ^ Low	mmol/L	meq/L	mg/dL
pH ^ Status _ mV [*]	pH ^ Status _ mV	mV		
pCO2 ^ Status _ mV [*]	pCO2 ^ Status _ mV [*]	mV		
K+ ^ Status _ mV [*]	K+ ^ Status _ mV [*]	mV		
Na+ ^ Status _ mV [*]	Na+ ^ Status _ mV [*]	mV		
Ca++ ^ Status _ mV [*]	Ca++ ^ Status _ mV [*]	mV		
Cl- ^ Status _ mV [*]	Cl- ^ Status _ mV [*]	mV		
pO2 ^ Sens [*]	pO2 ^ Sens [*]	mV		
pO2 ^ Status [*]	pO2 ^ Status [*]	mV		
Hct ^ Sens	Hct ^ Sens			

^{*} Parameters and units are available on the ABL90 FLEX analyzer only

[†] Parameters and units are available on the ABL80 FLEX and ABL9 analyzers only

[‡] Parameters and units are not available on the ABL9 analyzer

[§] Parameters and units are not available on the ABL80 or ABL9 analyzers

[#] Parameters and units are available on the ABL80 FLEX analyzer only

**Patient
information
for the
RADIANCE
system**

The following table lists the allowable unit strings for patient information parameters contained in the P record or PID segment for the RADIANCE system.

Parameter	Field / Component	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Patient age	15/2	years	months	days	weeks	hours
Patient height	17/2	feet	cm	m	in	
Patient weight	18/2	kg	lbs	oz	g	
Patient gestational age	16/2	years	months	days	weeks	hours
Patient birth weight	16/4	kg	lbs	oz	g	

**Patient
information
for the
RMED
analyzer**

The following table lists the allowable unit strings for patient information parameters contained in the P record or PID segment for the RMED analyzer.

Parameter	Field / Component	Unit 1	Unit 2	Unit 3
Patient age	15/2	years	months	days
Patient height	17/2	feet	cm	m
Patient weight	18/2	kg	lbs	
Patient gestational age	16/2	weeks		
Patient birth weight	16/4	kg	oz	g

List of parameters and units for the AQT90 FLEX analyzer

Patient results

The following gives a list of parameters and unit names for patient results for the AQT90 FLEX analyzer.

Parameter	ASTM / HL7	ASTM / 6xx	Unit 1	Unit 2	Unit 3	Unit 4
TnI	TnI	TnI	µg/L	ng/mL	ng/L	pg/mL
TnT	TnT	TnT	µg/L	ng/mL	ng/L	pg/mL
CKMB	CKMB	CKMB	µg/L	ng/mL		
Myo	Myo	Myo	µg/L	ng/mL		
BNP	BNP	BNP	ng/L	pg/mL		
NT-proBNP	NT-proBNP	NT-proBNP	ng/L	pg/mL		
D-dimer	D-dimer	D-dimer	µg/L	ng/mL	mg/L	
CRP	CRP	CRP	mg/L	µg/mL	mg/dL	
βhCG	beta-hCG	beta-hCG	IU/L	mIU/mL		
hsCRP	hsCRP	hsCRP	mg/L	µg/mL	Mg/dL	
PT-INR	PT-INR	PT-INR				

Extended handshake options

New options When requesting information from a HIS/LIS, the ASTM **Q** record may have the **Request Information Status Code** field set as follows:

When querying information based on patient id, the code "D" is used

When querying information based on accession number, the code "O" is used

To enable the **Request Information Status Code**, the **Q** record line of the **Patient Information Query** section in the relevant protocol ".ini" file must be changed in the analyzer as follows (added text underlined):

```
Q=Q|1|PID1234862255^ACN2233445609||||||||D<CR>
```

For cancellation of an outstanding request by means of the **Request Information Status Code** value "A" before submission of a new query and when leaving the DMS "PatientID" form with an outstanding query the analyzer configuration has to be changed by a Radiometer technician.

Termination code The **Termination Code** of a reply to an information request may be checked to control if the reply should terminate an outstanding request. To enable this check contact a Radiometer technician.

Activity log messages

Introduction The following contains lists of system error messages, one for the ABL800 FLEX analyzer one for the AQT90 FLEX analyzer, one for the ABL90 FLEX analyzer, one for the ABL80 FLEX analyzer, and one for the ABL9 analyzer.

Activity log messages – ABL800 FLEX analyzer

NOTICE: The list may be updated during upgrades of the ABL800 FLEX software. To get an updated list contact your Radiometer representative.

ErrorCode	ErrorName
1	Inconsistent software versions. Please contact service.
83	Value above reference range
84	Value below reference range
85	Value is below the critical limit
86	Value is above the critical limit
89	Measured QC value above the control range
90	Measured QC value below control range
93	Value above the reportable range
94	Value below the reportable range
117	LIS/HIS: Invalid connection configuration
128	LIS/HIS: Failed to open connection
129	LIS/HIS: Failed to close connection
131	LIS/HIS: Failed to send packet
132	LIS/HIS: Failed to receive packet
133	LIS/HIS: Connection lost
134	LIS/HIS: Connection established
165	LIS/HIS: High level protocol could not generate high level packet

ErrorCode	ErrorName
166	LIS/HIS: General communication error
167	LIS/HIS: High level protocol received packet in wrong format
200	User msg:
201	Westgard Rule (1.2s) violation
202	Westgard Rule (1.3s) violation
203	Westgard Rule (2.2s) violation
204	Westgard Rule (R.4s) violation
205	Westgard Rule (4.1s) violation
206	Westgard Rule (10.x) violation
207	Calibration schedule reminder(s) present
208	Quality control schedule reminder(s) present
209	Replacement schedule reminder(s) present
210	Calibration error(s) present
211	Quality control error(s) present
212	System message(s) present
213	Automatic backup failed
214	Automatic backup succeeded
216	Printer error
217	Replacement:
290	Warning: SHb detected
291	SHb too high
292	Turbidity too high
293	Warning: HbF detected and compensated for
329	QC expiration date exceeded
331	No sample detected during sample aspiration
357	Temp. error: Barometer
375	Calibration status out of limits

ErrorCode	ErrorName
376	Calibration Drift 1 out of range
377	Calibration Drift 2 out of range
378	Calibration sensitivity out of range
379	Calibration unstable (response fault)
443	Ca(7.4) not usable
452	Interference during measurement
484	Today is last day in current statistical month - remember to print QC statistics
487	A new statistical month has begun - remember to export WDC data
494	Bilirubin too high
508	Rinse Error
512	Temperature error
521	Inhomogeneous sample
522	Calibration error
523	Calibration drift out of range
529	Inlet liquid sensor error
531	pH/BG upper Liquid Sensor error
537	OXI Liquid Sensor error
581	OXI spectrum mismatch
582	tHb calibration outside limits
584	Oximetry calibration error
588	Measured QC value lower than statistical range
589	Measured QC value higher than statistical range
593	Insufficient sample
595	Liquid sensor calibration error
606	Cal expired (pH)
608	Cal expired (pCO2)
609	Cal expired (pO2)

ErrorCode	ErrorName
610	Cal expired (K)
611	Cal expired (Na)
612	Cal expired (Ca)
613	Cal expired (Cl)
614	Cal expired (Glu)
615	Cal expired (Lac)
616	Cal expired (OXI)
641	ABL/DMS PC restarted
642	ABL/DMS PC Connected to AU
643	ABL/DMS PC Disconnected from AU
648	Calibration failed or not accepted
662	Barometer out of range
669	QC value outside control range
679	Barometer error
682	OXI module not active
688	ctHb/ceHb too low for OXI calculation
692	ABL not connected to RADIANCE
693	ABL not connected to RADIANCE - incorrect password
694	ABL connected to RADIANCE
695	ABL disconnected from RADIANCE
696	ABL<>RADIANCE communication error
699	AutoCheck QC measurement started due to calibration error
700	Scheduled AutoCheck not run due to errors in last calibration
703	QC expired
704	AutoCheck QC measurement is repeated
705	AutoCheck QC measurement is repeated twice

ErrorCode	ErrorName
707	Replacement(s) overdue by 10%. Analyzer locked.
708	Corrective action not possible due to empty AutoCheck carousel
712	FHbF measurement not possible
713	ctBil measurement not possible
734	DMS not connected to Analyzing Unit
745	Low hard disk space
766	ABL not connected to RADIANCE - no RADIANCE connection license
767	ABL not connected to RADIANCE - ABL StatLink version too high
768	ABL not connected to RADIANCE - ABL StatLink version too low
769	ABL<>RADIANCE communication error - XML packet could not be parsed
770	Failed to restore Custom Setup
771	Succeeded to restore Custom Setup
772	User Activity:
773	Remote operator logged on with user:
774	Remote operator logged off with user:
775	Failed to restore Default Setup
776	Succeeded to restore Default Setup
780	RADIANCE communication enabled
781	RADIANCE communication disabled
782	RADIANCE output queue cleared
783	Automatic backup started
785	Automatic archiving started
786	Automatic archiving completed
787	Export of data logs started

ErrorCode	ErrorName
798	User logged on
799	User logged off
800	Logon attempt failed
810	pH locked
811	pCO2 locked
812	pO2 locked
813	K locked
814	Na locked
815	Cl locked
816	Ca locked
818	Glu locked
819	Lac locked
820	tHb locked
821	MetHb locked
822	COHb locked
823	HHb locked
824	O2Hb locked
825	sO2 locked
826	HbF locked
827	tBil locked
831	pH unlocked
832	pCO2 unlocked
833	pO2 unlocked
834	K unlocked
835	Na unlocked
836	Cl unlocked
837	Ca unlocked
839	Glu unlocked

ErrorCode	ErrorName
840	Lac unlocked
841	tHb unlocked
842	MetHb unlocked
843	COHb unlocked
844	HHb unlocked
845	O2Hb unlocked
846	sO2 unlocked
847	HbF unlocked
848	tBil unlocked
852	RADIANCE:
855	Base Excess out of range
875	Sample aged
885	Cyclic QC schedule reset from RADIANCE
886	LIS/HIS: No valid POCT1A DML Device ID file
963	Leak currents detected
964	Leak currents detected
979	Inhomogeneous rinse solution
980	Inhomogeneous QC1 solution

Activity log messages – AQT90 FLEX analyzer

ErrorCode	ErrorName
1	Inconsistent software versions. Please contact service.
83	Value above reference range
84	Value below reference range
85	Value is below the critical limit
86	Value is above the critical limit
89	Measured QC value above the control range
90	Measured QC value below control range
93	Value above the reportable range
94	Value below the reportable range
99	LIS/HIS: Low level protocol send failed
100	LIS/HIS: Low level protocol receive failed
101	LIS/HIS: Low level protocol unable to send because connection is busy
102	LIS/HIS: Low level protocol receive failed because of timeout
109	LIS/HIS: Low level protocol send failed, ACK or NAK not received
115	LIS/HIS: Low level protocol ACK after ENQ not received
117	LIS/HIS: Invalid connection configuration
118	LIS/HIS: Connection manager could not open high level protocol
128	LIS/HIS: Failed to open connection
129	LIS/HIS: Failed to close connection
131	LIS/HIS: Failed to send packet
132	LIS/HIS: Failed to receive packet

ErrorCode	ErrorName
133	LIS/HIS: Connection lost
134	LIS/HIS: Connection established
135	LIS/HIS: High level protocol could not open connection
141	LIS/HIS: Low level protocol configuration not correct
142	LIS/HIS: Low level protocol failed to open connection
143	LIS/HIS: Low level protocol failed to close connection
165	LIS/HIS: High level protocol could not generate high level packet
166	LIS/HIS: General communication error
167	LIS/HIS: High level protocol received packet in wrong format
200	User msg:
201	Westgard Rule (1.2s) violation
202	Westgard Rule (1.3s) violation
203	Westgard Rule (2.2s) violation
204	Westgard Rule (R.4s) violation
205	Westgard Rule (4.1s) violation
206	Westgard Rule (10.x) violation
207	Calibration schedule reminder(s) present
208	Quality control schedule reminder(s) present
209	Replacement schedule reminder(s) present
210	Calibration error(s) present
211	Quality control error(s) present
212	System message(s) present
213	Automatic backup failed
214	Automatic backup succeeded
215	Automatic backup failed. Error code:

ErrorCode	ErrorName
216	Printer error
217	Replacement:
328	No leading air segment in inlet's liquid sensor within time frame
329	QC expiration date exceeded
476	Measurement unstable
484	Today is last day in current statistical month - remember to print QC statistics
487	A new statistical month has begun - remember to export WDC data
508	Rinse Error
512	Temperature error
514	Communication error
516	Bottle empty or transport error
521	Inhomogeneous sample
522	Calibration error
523	Calibration drift out of range
524	Measuring error
525	Conditioning error
545	General communication error
547	Temperature error
574	Missing Cal data
576	Pump calibration error
577	License key missing
578	Option management error
583	Measured value outside reportable range for the parameter
587	Option key error
588	Measured QC value lower than statistical range

ErrorCode	ErrorName
589	Measured QC value higher than statistical range
593	Insufficient sample
595	Liquid sensor calibration error
600	Demo option will expire soon
603	Options being installed
604	Parameter not installed
605	Cal expired
618	Waste detector error
619	Waste container missing
620	Waste container full
621	Waste container critically full
622	Expected liquid level below 0
623	Solution empty
624	Solution missing
641	ABL/DMS PC restarted
642	ABL/DMS PC Connected to AU
643	ABL/DMS PC Disconnected from AU
648	Calibration failed or not accepted
659	Cover removed or window opened
660	Inlet calibration error
661	Leak detected
662	Barometer out of range
663	Leakage
664	Sample problem
669	QC value outside control range
671	Analyzing Unit service setup non-default
672	Inlet error: Pickup offset adjust error
673	Inlet error: Optic ref point error

ErrorCode	ErrorName
674	Inlet error: No gasket detected
675	Inlet error: Position calc. error
676	Inlet error: Sensitivity out of range
677	Unreliable
678	Heater error
679	Barometer error
683	Inlet module not active
686	Cal 2 Solution missing - check level in container
687	Cal 1 Solution missing - check level in container
689	Gas pressure offset voltage out of range.
690	Automatic AU SW Repair
691	AU in bootstrap mode
692	ABL not connected to RADIANCE
693	ABL not connected to RADIANCE - incorrect password
694	ABL connected to RADIANCE
695	ABL disconnected from RADIANCE
696	ABL<>RADIANCE communication error
703	QC expired
707	Replacement(s) overdue by 10%. Analyzer locked.
722	Sample error
723	No Cleaning solution detected by Inlet during Startup
724	No Cal 2 solution detected by Inlet during Startup
725	No Cal 1 solution detected by Inlet during Startup
726	No Rinse solution detected by Inlet during Startup
727	No Cleaning solution detected by Inlet during Refill
728	No Cal 2 solution detected by Inlet during Refill
729	No Cal 1 solution detected by Inlet during Refill

ErrorCode	ErrorName
730	No Rinse solution detected by Inlet during Refill
733	New part installed
734	DMS not connected to Analyzing Unit
735	Ampoule temperature error
737	Calibration solution missing. Check level in container
739	Solution's composition not identifiable
745	Low hard disk space
746	Rinse Solution low warning
747	Cal1 Solution low warning
748	Cal2 Solution low warning
749	Cleaning Solution low warning
750	Solution projection adjusted
752	Cal3 Solution low warning
753	Service Electrode Updatings enabled
765	Inlet flow impeded
766	ABL not connected to RADIANCE - no RADIANCE connection license
767	ABL not connected to RADIANCE - ABL StatLink version too high
768	ABL not connected to RADIANCE - ABL StatLink version too low
769	ABL<>RADIANCE communication error - XML packet could not be parsed
770	Failed to restore Custom Setup
771	Succeded to restore Custom Setup
772	User Activity:
773	Remote operator logged on with user:
774	Remote operator logged off with user:
775	Failed to restore Default Setup

ErrorCode	ErrorName
776	Succeeded to restore Default Setup
777	Database upgrade running in background
778	Inlet gasket should be replaced soon
779	Cleaning failed
780	RADIANCE communication enabled
781	RADIANCE communication disabled
782	RADIANCE output queue cleared
783	Automatic backup started
785	Automatic archiving started
786	Automatic archiving completed
787	Export of data logs started
788	Hypochlorite detected - remembrane Glu, Lac and Cl electrodes
790	Adaptive measuring mode applied
791	Parameter could not be measured
796	Solution pump asymmetry error
797	Waste pump asymmetry error
798	User logged on
799	User logged off
800	Logon attempt failed
801	Hard disk S.M.A.R.T. threshold(s) exceeded
807	Unauthorized software
808	Calibration drift not available
809	Service Electrode Updatings disabled
852	RADIANCE:
853	LIS/HIS: Missing Communication Option Key
854	No leading air segment in inlet's lower liquid sensor within time frame
875	Sample aged

ErrorCode	ErrorName
876	FLEXQ positioning warning
877	FLEXQ positioning error
878	No Rinse solution during calibration
879	No Cal1 solution during calibration
880	No Cal2 solution during calibration
881	LIS/HIS: Missing CIC Communication Option Key
882	Mixer tray slip detected
883	Mixer error
884	Sampler tray adjustment needed
885	Cyclic QC schedule reset from RADIANCE
886	LIS/HIS: No valid POCT1A DML Device ID file
887	ReplenishPack capacity low
888	DILUTION capacity low
889	HCG_BETA capacity low
890	TROPONIN_I capacity low
891	CKMB capacity low
892	D_DIMER capacity low
893	USCRP capacity low
894	MYOGLOBIN capacity low
895	NT capacity low
896	Calibration job submission failed
897	Calibration verified
898	Calibration verification invalidated
899	Calibration verification expired
900	Calibration verification is about to expire
901	Less than five percent free disk space remaining
902	Less than fifteen percent free disk space remaining
903	Automatic Backup Setup changed

ErrorCode	ErrorName
907	Vacutainer cap color was not read by the Analyzer, but selected by the operator.
1000	Needle Clean Activity Completed
1001	Needle Change Activity Completed
1002	Service Inspection Activity Completed
1003	Needle Clean Counter Exceeds Warning Limit
1004	Needle Clean Counter Exceeds CriticalLimit
1005	Needle Change Counter Exceeds Warning Limit
1006	Needle Change Counter Exceeds Critical Limit
1007	Needle Clean Scheduled time exceeds
1008	Service Inspection Scheduled time exceeds
1009	User Activity Scheduled time exceeds
1010	Needle Clean Activity has Failed
1011	Needle Clean Activity has Started
1020	Incubation Temperature Out of Range
1021	Analyzer has low flash disk spare blocks
1022	Calibration task Interrupted
1023	Calibration task aborted
1024	Calibration Blank_outliers check failed
1025	Calibration Calibrator_outliers check failed
1026	The amount of counts measured per cycle above the valid limit
1027	Incubation Time Out of Range
1028	Maximum Dark Measurement Difference out of Range
1029	Difference between Measured value and reference value out of Range
1030	Dry Temperature out of range
1031	Integration check for Measured value failed

ErrorCode	ErrorName
1032	Shake Time Out of Range
1033	Incubation temperature standard deviation Out of Range
1034	AS Software version consistence check failed
1035	Maximum Dark measurements check failed
1036	Volume in the test cup out of range
1037	Reference sample time constant out of range
1038	Mean Incubation temperature out of range
1039	HCG_BETA LOCKED
1040	TROPONIN_I LOCKED
1041	CK_MB LOCKED
1042	D_DIMER LOCKED
1043	USCRP LOCKED
1044	MYOGLOBIN LOCKED
1045	HCG_BETA UNLOCKED
1046	TROPONIN_I UNLOCKED
1047	CK_MB UNLOCKED
1048	D_DIMER UNLOCKED
1049	USCRP UNLOCKED
1050	MYOGLOBIN UNLOCKED
1051	No Test available for measurement
1052	Replenish Pack not accessible
1053	Replenish Pack expires with in threshold time
1054	Calibration Failed for @x with LotNo @y
1055	Calibration Interrupted for @x with LotNo @y
1056	Calibration Aborted for @x with LotNo @y
1057	Needle Pierce count exceeded
1058	Dilution cup volume out of range

ErrorCode	ErrorName
1059	Calibration Process Failed
1060	Calibration Process Interrupted
1061	Calibration Process Aborted
1062	Sample Registration interrupted by user

Activity log messages – ABL90 FLEX analyzer

NOTICE: The list may be updated during upgrades of the analyzer software. To get an updated list contact your Radiometer representative.

ErrorCode	ErrorMsg
1	Inconsistent software versions. Please contact service
83	Value above reference range
84	Value below reference range
85	Value below critical limit
86	Value above critical limit
89	Measured QC value above control range
90	Measured QC value below control range
93	Value above reportable range
94	Value below reportable range
117	LIS/HIS: Invalid connection configuration
128	LIS/HIS: Failed to open connection
129	LIS/HIS: Failed to close connection
131	LIS/HIS: Failed to send packet
132	LIS/HIS: Failed to receive packet
133	LIS/HIS: Connection lost
134	LIS/HIS: Connection established
165	LIS/HIS: High-level protocol could not generate high-level packet
166	LIS/HIS: General communication error
167	LIS/HIS: High-level protocol received packet in wrong format
200	User msg:
201	Westgard Rule (1.2s) violation
202	Westgard Rule (1.3s) violation
203	Westgard Rule (2.2s) violation

ErrorCode	ErrorMsg
204	Westgard Rule (R.4s) violation
205	Westgard Rule (4.1s) violation
206	Westgard Rule (10.x) violation
207	Calibration schedule reminder(s) present
208	Quality control schedule reminder(s) present
209	Replacement schedule reminder(s) present
210	Calibration error(s) present
211	Quality control error(s) present
212	System message(s) present
213	Automatic backup failed
214	Automatic backup succeeded
216	General printer error
217	Replacement:
290	Warning: SHb detected
291	SHb too high
292	Turbidity too high
293	Oxi compensated for HbF
329	QC expiration date exceeded
331	No sample detected during sample aspiration
357	Temp. error: Barometer
375	Calibration status out of limits
376	Calibration Drift 1 out of range
377	Calibration Drift 2 out of range
378	Calibration sensitivity out of range
379	Calibration unstable (response fault)
408	Zero current error
443	Ca(7.4) not usable
452	Interference during measurement

ErrorCode	ErrorMsg
484	Today is last day in stat. month - remember to print QC statistics
487	A new statistical month has begun - remember to export WDC data
494	Bilirubin too high
508	Rinse error
512	Temperature error
521	Inhomogeneous sample
522	Calibration error
523	Calibration drift out of range
529	Inlet LS failed to calibrate
531	Sensors LS failed to calibrate
537	OXI LS failed to calibrate
576	Pump calibration error
577	License key missing
578	Option management error
581	OXI spectrum mismatch
582	tHb calibration cuvette factor outside limits
584	tHb calibration wavelength outside limits
588	Measured QC value lower than statistical range
589	Measured QC value higher than statistical range
593	Insufficient sample
595	Liquid sensor calibration error
600	Demo option will expire soon
604	Parameter not installed
606	Cal expired (pH)
608	Cal expired (pCO2)
609	Cal expired (pO2)
610	Cal expired (K)

ErrorCode	ErrorMsg
611	Cal expired (Na)
612	Cal expired (Ca)
613	Cal expired (Cl)
614	Cal expired (Glu)
615	Cal expired (Lac)
616	Cal expired (OXI)
641	ABL/DMS PC restarted
642	ABL/DMS PC connected to wet section
643	ABL/DMS PC disconnected from wet section
648	Calibration failed or not accepted
662	Barometer out of range
679	Barometer error
682	OXI module not active
688	ctHb/ceHb too low for OXI calculation
692	ABL not connected to RADIANCE
693	ABL not connected to RADIANCE - incorrect password
694	ABL connected to RADIANCE
695	ABL disconnected from RADIANCE
696	ABL<>RADIANCE communication error
699	Built-in QC measurement started due to calibration error
700	Scheduled built-in QC not run due to errors in last calibration
703	QC expired
704	Built-in QC measurement is repeated
705	Built-in QC measurement is repeated twice
707	Replacement(s) overdue by 10 %. Analyzer locked.
708	Corrective action not possible due to empty solution pack

ErrorCode	ErrorMsg
712	FHbF measurement not possible
713	ctBil measurement not possible
733	New part installed
734	General WSM exception
745	Low disk space
766	ABL not connected to RADIANCE - no RADIANCE connection license
767	ABL not connected to RADIANCE - ABL StatLink version too high
768	ABL not connected to RADIANCE - ABL StatLink version too low
769	ABL<>RADIANCE communication error - XML packet could not be parsed
770	Failed to restore Custom Setup
771	Succeeded to restore Custom Setup
772	User Activity:
773	Remote operator logged on with user:
774	Remote operator logged off with user:
775	Failed to restore Default Setup
776	Succeeded to restore Default Setup
780	RADIANCE communication enabled
781	RADIANCE communication disabled
782	RADIANCE output queue cleared
783	Automatic backup started
785	Automatic archiving started
786	Automatic archiving completed
787	Export of data logs started
798	User logged on
799	User logged off

ErrorCode	ErrorMsg
800	Logon attempt failed
810	pH locked
811	pCO2 locked
812	pO2 locked
813	K locked
814	Na locked
815	Cl locked
816	Ca locked
818	Glu locked
819	Lac locked
820	tHb locked
821	MetHb locked
822	COHb locked
823	HHb locked
824	O2Hb locked
825	sO2 locked
826	HbF locked
827	tBil locked
831	pH unlocked
832	pCO2 unlocked
833	pO2 unlocked
834	K unlocked
835	Na unlocked
836	Cl unlocked
837	Ca unlocked
839	Glu unlocked
840	Lac unlocked
841	tHb unlocked

ErrorCode	ErrorMsg
842	MetHb unlocked
843	COHb unlocked
844	HHb unlocked
845	O2Hb unlocked
846	sO2 unlocked
847	HbF unlocked
848	tBil unlocked
852	RADIANCE:
853	LIS/HIS: Missing Communication Option Key
855	Base Excess out of range
875	Sample aged
881	LIS/HIS: Missing CIC Communication Option Key
885	Cyclic QC schedule reset from RADIANCE
886	LIS/HIS: No valid POCT1A DML Device ID file
963	Leak currents detected
964	Leak currents detected
969	Sensor conditioning started
970	Replace solution pack
971	Replace sensor cassette
972	Sensor conditioning due
973	Printer paper must be replaced
978	Flow selector calibration error
979	Inhomogeneous rinse solution
980	Inhomogeneous QC1 solution
981	Inhomogeneous QC2 solution
982	Inhomogeneous QC3 solution
983	Inhomogeneous cal 3 solution
984	The analyzer could not aspirate homogeneous

ErrorCode	ErrorMsg
	calibration solution
1000	Number of pO2 hardware data fail
1001	Timeout while waiting for pO2 hardware data
1002	pO2 dark data is out of range
1004	Unable to calculate oxygen parameter
1005	Unable to calculate oxygen parameter
1006	Unable to calculate oxygen parameter
1007	Missing oxygen calibration
1008	Unable to calculate oxygen parameter
1009	Unable to calculate oxygen parameter
1010	Oxi data collection error
1011	Oxi has no Blank Cal
1012	Oxi has no sample spectrum
1013	Oxi data collection error
1014	Oxi Blank Cal. intensity too high
1015	Oxi sample intensity too high
1016	Oxi Blank Cal. intensity too low
1017	Oxi sample intensity too low
1018	Oxi electronic adjustment error
1019	Oxi Blank Cal. outside limits
1020	Oxi neon intensity outside limits
1021	Oxi neon correction outside limits
1022	Oxi background correction outside limits
1023	Oxi spectrometer memory read problem
1024	Oxi spectrometer memory write problem
1025	Oxi hemolyzer tuning problem
1026	Oxi hemolyzer frequency problem
1027	Oxi hemolyzer temperature deviation too high

ErrorCode	ErrorMsg
1028	Oxi neon voltage outside limits
1029	Oxi light source voltage outside limits
1030	Oxi hemolyzer voltage outside limits
1031	Oxi initialization in progress
1032	Oxi data collection problem
1033	Oxi task was not finished
1034	Oxi hardware problem
1045	Unable to read consumable information
1046	Error in communication with one-wire chip
1047	Error in communication with one-wire chip
1048	Error in communication with one-wire chip
1049	Error in communication with one-wire chip
1050	Error in communication with one-wire chip
1051	Error in communication with one-wire chip
1052	Error in communication with one-wire chip
1053	Error in communication with one-wire chip
1054	Error in communication with one-wire chip
1055	Error in communication with one-wire chip
1056	Error in communication with one-wire chip
1057	Error in communication with one-wire chip
1058	Error in communication with one-wire chip
1059	Error in communication with one-wire chip
1060	Error in communication with one-wire chip
1061	Pressure test flow error
1062	Pressure test pressure error
1063	Pressure test vacuum error
1064	Temperature in sensor cassette top out of range
1065	Temperature in sensor cassette bottom out of range

ErrorCode	ErrorMsg
1066	Temperature in sensor cassette substrate out of range
1069	Temperature in Oxi cuvette out of range
1071	Temperature in Oxi spectrometer out of range
1080	Conditioning changed
1081	Inhomogeneous rinse solution
1083	Inhomogeneous cal 2 solution
1084	Inhomogeneous cal 3 solution
1085	Inhomogeneous QC1 solution
1086	Inhomogeneous QC2 solution
1087	Inhomogeneous QC3 solution
1088	Inhomogeneous cleaning solution
1089	Inhomogeneous gas
1090	No rinse solution
1092	No cal 2 solution
1093	No cal 3 solution
1094	No QC1 solution
1095	No QC2 solution
1096	No QC3 solution
1097	No cleaning solution
1098	No gas
1099	Pump calibration error
1100	Outlet LS not empty during pump calibration
1101	Outlet LS not full during pump calibration
1108	Full air not detected
1109	Inhomogeneous full air detected
1110	No air detected
1111	Inhomogeneous air

ErrorCode	ErrorMsg
1112	LS inlet not empty
1113	LS sensors not empty
1114	LS outlet not empty
1115	Ws communication error: wrong message format
1116	Ws communication error: keep alive timeout
1117	Oxi spectrometer temperature drift
1120	Sensor replacement successful
1121	The port did not open during sensor replacement
1123	The sensor chip data could not be read or written during replacement
1124	An unregistered sensor was installed during replacement
1125	An unregistered and used sensor was installed during replacement
1126	A registered sensor had been used before installation
1134	The chip information for the solution pack cannot be read or written
1135	The solution pack has been used before
1140	The solution pack has used the maximum number of measurements at installation
1142	The printer door is open. Printing not possible
1143	Internal printer is offline. Printing not possible
1144	Check that printer door is closed and that paper is present
1145	A printer error has occurred. Call service technician
1146	Printer paper replaced
1147	Inlet opened during rinse
1148	Inlet open during calibration
1149	Inlet open during wet section activity
1150	Inlet closed without aspirating sample

ErrorCode	ErrorMsg
1151	Inlet not closed: no sample aspirated
1152	Could not read or write to chip during replacement
1157	No valid FTC programs detected
1165	Solution pack not properly installed
1166	Solution pack expired
1167	Sensor cassette not properly installed
1168	Sensor cassette expired
1169	Unable to pump solutions
1170	Inlet has been open for too long
1171	Inlet is missing or in unknown state
1172	Sensor cassette damaged
1173	Solution pack damaged
1174	Inlet opened while the analyzer was busy
1175	Sensor temperature error
1176	A liquid sensor error was detected
1177	A flow selector error was detected
1178	A pump calibration error was detected
1179	Currently installed option keys do not match hardware
1180	An error occurred when trying to communicate with wet section
1181	A software or hardware error exists in wet section
1183	Valve malfunctioning
1184	Leak detected
1185	Warning: Free memory is low
1186	Free system memory is critically low
1187	Disk shows signs of wear
1188	Disk shows serious signs of wear
1189	FTC aborted, LS state change error

ErrorCode	ErrorMsg
1190	Inlet open
1191	QA Portal communication enabled
1192	QA Portal communication disabled
1193	QA Portal output queue cleared
1194	ABL not connected to QA Portal
1195	ABL not connected to QA Portal - incorrect password
1196	QA Portal data transmission successful
1197	QA Portal data transmission failed
1198	ABL<>QA Portal communication error - XML packet could not be parsed
1199	FTC program has been retried
1200	Solution pack empty
1201	Solution pack lifetime expired
1202	Expiration date reached
1219	RiLiBÄK Violation: Value above upper limit
1220	RiLiBÄK Violation: Value below lower limit
1221	System temperature out of range
1222	Temperature system error
1223	Analyzer did not connect at start-up
1224	Analyzer is temporarily shut down
1225	The sample is older than a day
1226	The sample age is negative
1227	Correction for bicarbonate contains errors from pH, pCO ₂
1228	Correction for lactate contains errors from K ⁺ , Na ⁺ , Ca ²⁺
1230	Inlet gasket replaced
1231	Inlet probe replaced
1232	Inlet connection gasket replaced

ErrorCode	ErrorMsg
1233	Inlet cleaned
1295	Activity has been repeated due to the following reason:
1296	Printer out of paper
1297	Printer is offline
1298	Printer lid open
1299	Rinse activity repeated:
1300	Calibration activity repeated:
1301	QC activity repeated:
1302	Startup activity repeated:
1303	Activity repeated:
1304	Calibration activity repeated
1305	End of repeat reason list
1306	Solution pack manually removed
1307	Disk space less than fifteen percent
1308	Disk space less than five percent
1310	Response error

Activity log messages – ABL80 FLEX analyzers

NOTICE: ABL80 FLEX software utilizes the activity log message with either Standard or Replacement types to transmit the actual event logs to the RADIANCE system. This activity log message is not sent to HISLIS.

Event	User messages	Comments
System Cycle	System Cycle (Auto)	This event is specific to ABL80 FLEX, ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers.
	System Cycle (Manual)	
	System Cycle (Remote)	
	System Cycle (Init)	
	System Cycle (Reinit)	
	System Cycle (Mandatory)	
	System Cycle (Startup)	
	System Cycle (C/A)	
	System Cycle (V/C)	
	System Cycle (no V/C)	
	System Cycle (Init V/C)	
	System Cycle (Init no V/C)	
	System Cycle (Reinit V/C)	
	System Cycle (Reinit no V/C)	
Two Point Calibration	2 pt Cal (Auto)	
	2 pt Cal (Manual)	
	2 pt Cal (Remote)	
	2 pt Cal (Init)	
	2 pt Cal (Reinit)	
	2 pt Cal (Mandatory)	
	2 pt Cal (Startup)	

Event	User messages	Comments
	2 pt Cal (V/C)	
	2 pt Cal (no V/C)	
	2 pt Cal (Init V/C)	
	2 pt Cal (Init no V/C)	
	2 pt Cal (Reinit V/C)	
	2 pt Cal (C/A)	
CO-OX Blank Calibration	CO-OX blank cal (Auto)	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers
	CO-OX blank cal (Manual)	
	CO-OX blank cal (Remote)	
	CO-OX blank cal (Init)	
	CO-OX blank cal (Reinit)	
	CO-OX blank cal (Mandatory)	
	CO-OX blank cal (Startup)	
	CO-OX blank cal (V/C)	
	CO-OX blank cal (no V/C)	
	CO-OX blank cal (Init V/C)	
	CO-OX blank cal (Init no V/C)	
	CO-OX blank cal (Reinit V/C)	
	CO-OX blank cal (Reinit no V/C)	
	CO-OX blank cal (tHb Cal)	
	CO-OX blank cal (C/A)	
Liquid Sensor Calibration	Liquid Sensor Calibration	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers

Event	User messages	Comments
tHb Calibration	tHb Calibration	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers
System Check	System check	
Patient Analysis	Patient	
Manual QC Analysis	Manual QC (L1)	
	Manual QC (L2)	
	Manual QC (L3)	
	Manual QC (L4)	
	Manual QC (Hct L1)	
	Manual QC (Hct L2)	
Inactivate Parameter(s) on Current SC	Inactivate (current SC)	
Inactivate Parameter(s) on All SC	Inactivate (all SC)	
Activate Parameter(s) on Current SC	Activate (current SC)	
Activate Parameter(s) on All SC	Activate (all SC)	
Inactivate Air Detection	Air detection inactivated	This event does not apply to the ABL80 FLEX CO-OX with OSM software configuration analyzer
Activate Air Detection	Air detection activated	This event does not apply to the ABL80

Event	User messages	Comments
		FLEX CO-OX with OSM software configuration analyzer
Analyzer Lockout	Analyzer locked	
	Analyzer unlocked	
Parameter Lockout	Parameter locked	
	Parameter unlocked	
Database Compression	DB compression	
Install Solution Pack	SP install	
Install Sensor Cassette	SC install	
Reinstall Sensor Cassette	SC reinstall	
CO-OX Temperature Error	CO-OX temp error	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers
CO-OX Board Failure	CO-OX control board failure	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers
CO-OX Fluidic Pathway Check	Fluidic Pathway Check	
CO-OX Fluidic Pathway Blockage	Blockage	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX

Event	User messages	Comments
		CO-OX with OSM software configuration analyzers
CO-OX Liquid Sensor	Liquid sensor	This event is specific to ABL80 FLEX CO-OX and ABL80 FLEX CO-OX with OSM software configuration analyzers
Standby Mode	Standby (Enter)	This event is specific to ABL80 FLEX analyzers with BASIC software configuration and ABL80 FLEX CO-OX with OSM software configuration analyzers
	Standby (Exit)	
	Standby (Flush)	
	Standby (Exit Flush)	
SC Check	SC Check	This event is specific to ABL80 FLEX CO-OX with OSM software configuration analyzers
	SC Check (Init)	
	SC Check (Reinit)	
	SC Check (Init no V/C)	
	SC Check (Reinit no V/C)	
Internal QC	Internal QC (L1 Manual)	This event is specific to ABL80 FLEX CO-OX analyser with RiliBAK software configuration
	Internal QC (L2 Manual)	
	Internal QC (L1 Remote)	
	Internal QC (L2 Remote)	
	Internal QC (L1 Automatic)	
	Internal QC (L2 Automatic)	
	Internal QC (L1 Pending)	
	Internal QC (L2 Pending)	

Event	User messages	Comments
Cleaning	Cleaning	
Invalid User	Invalid logon BC	
Shutdown	System shutdown	

Activity log messages – ABL9 analyzer

NOTICE: There can be more than one message associated with some error codes.

Error code	Description
9001	Analyzer memory chip not found. Contact Service.
9002	Analyzer memory chip corrupt. Contact Service.
9003	Analyzer memory chip corrupt. Contact Service.
9004	Analyzer memory chip corrupt. Contact Service.
9005	Heater memory chip not detected. Contact Service.
9006	Calibration memory chip not detected. Contact Service.
9007	Analyzer memory chip corrupt. Contact Service.
9008	Interface board not found. Contact Service.
9009	Analyzer memory chip corrupt. Contact Service.
9010	Temperature threshold exceeded. Analyzer was shut down. Contact Service.
9011	Interface board communication lost during analysis. Unable to provide sample results.
9012	Analyzer memory ship not detected. Contact Service.
9013	No parameters programmed on analyzer memory chip. Contact Service.
9014	Unable to update firmware version. The interface board is not responding. Contact Service.
9015	Unable to update firmware version. The system could not write to the interface board.
9016	Unable to update firmware version. The bin file is missing. Contact Service.
9018	System log records at maximum capacity
9101	SP memory chip is corrupt
9102	SP door was opened during analysis. Analysis has been cancelled.

9103	SP not connected during System Check.
9104	SP cannot be used with this software version
9105	SP memory chip is corrupt.
9107	SP memory chip is corrupt
9108	SP memory chip corrupt
9109	SP cannot be used with this analyzer configuration
9110	SP cannot be installed more than once
9111	SP was previously replaced on this analyzer and cannot be re-installed
9112	Current date in analyzer is prior to SP manufacture date. Verify analyzer date is correct.
9113	SP shelf-life is expired
9114	SP solutions too low to complete installation
9115	SP memory chip corrupt
9117	SP in-use life is expired
9118	SP is empty No solutions remaining
9120	SP is not installed
9122	SP has low number of tests remaining
9123	SP in-use life will expire in X days
9124	SP is not connected
9125	SP has insufficient solution to perform calibration
9126	SP verification check failed
9127	SP verification check failed No endpoints detected
9128	Verification check failed: acceptance criteria failed.
9129	SP is invalid due to date/time adjustment
9201	SC door was opened during analysis. Analysis has been cancelled.

9202	SC not connected during System Check.
9203	SC cannot be used with this software version
9206	SC memory chip is corrupt
9207	SC memory chip corrupt
9208	SC memory chip is corrupt
9209	SC memory chip corrupt
9210	SC cannot be used with this analyzer configuration
9211	SC shelf-life is expired
9212	SC in-use life is expired
9213	DEMO SC DETECTED
9214	SC memory chip corrupt
9215	SC memory chip is corrupt
9216	SC cannot be installed
9218	SC has no tests remaining
9219	SC has no available parameters
9220	SC is not installed
9221	SC has low number of tests remaining
9222	SC in-use life will expire in X days
9223	SC is not connected
9225	Description: SC inlet opened during Calibration.
9226	Demo Mode SC
9227	Not a Demo Mode SC
9228	SC is invalid due to date/time adjustment
9235	Current date in analyzer is prior to SC manufacture date. Verify analyzer date is correct.
9302	SC temperature high. Temperature reading was {x}.
9303	SC temperature low. Temperature reading was {x}.

9304	SC temperature malfunction. Temperature reading was {x}.
9315	System check records at maximum capacity
9339	Calibration records at maximum capacity
9340	Calibration failed
9341	All parameters failed calibration
9350	Air detection calibration failed
9351	Air was detected in sample.
9352	Air was detected in flush solution
9400	Sample aspirated was not a QC solution. Measurement results discarded.
9401	No parameters available for analysis.
9402	Manual QC level {X} lot will expire on {Date}
9403	Manual QC Level 1 is overdue
9404	Manual QC Level 2 is overdue
9405	Manual QC Level 3 is overdue
9406	Manual QC Level 4 is overdue
9407	Manual QC Hct Level 1 is overdue
9408	Manual QC Hct Level 2 is overdue
9409	Manual QC records at maximum capacity
9410	Manual QC entries is approaching its maximum limit
9411	Manual QC Level 1 is expired
9412	Manual QC Level 2 is expired
9413	Manual QC Level 3 is expired
9414	Manual QC Level 4 is expired
9415	Manual QC Hct Level 1 is expired
9416	Manual QC Hct Level 2 is expired
9501	Inlet not closed within required time limit. Sample will be discarded.
9502	Aspiration time limit exceeded. Analysis has been cancelled.

9600	Analysis cancelled by operator.
9601	Patient Analysis records at maximum capacity
9602	NO REPORTABLE PARAMETERS No parameters available for analysis
9603	STANDBY MODE System will exit Standby on {date time format}
9604	Rinse failed
9605	Analysis is disabled Current operator's security access does not allow sample analysis.
9606	System logging is enabled. To disable, proceed to the Logging setup screen.
9607	Software upgrade version {x.xx.x} is available.
9608	Software upgrade is complete
9609	Software upgrade failed
9612	Patient Analysis entries is approaching maximum limit
9613	Analysis is disabled Log on to perform sample analysis
9615	Demo Mode
9616	Software upgrade files are invalid
9617	System running low on disk space. Contact service.

Extended system messages

Optional comment

An optional ASTM C record or HL7 NTE segment may be added to a system message transfer in order to include the content of a user message, the item of a replacement and the type of an analyzer activity. To enable this additional data transfer, please contact your local Radiometer technician.

NOTICE:

The added C record is only transmitted when extra information (information in addition to the error code) is available. Error code 751 is used for analyzer activities, code 200 is used for user messages and code 217 is used for a replacement.

The following example shows the transmission of a message concerning replacement of a Na membrane:

```
H|\^&|||ABL735^|||||||1|20001026133628
```

```
P|1
```

```
O|1||Error
```

```
R|1||217|||||||20001026133212
```

```
C|1|I|Na Membrane|G
```

```
L|1|N
```

Additional input parameters

Introduction The following gives details about additional parameters that can be used.

Parameter	ASTM6xx	ASTM / HL7
Spontaneous RR	Spontaneous_RR	Spontaneous_RR
Set RR	Set_RR	Set_RR
Vt	Vt	Vt
Ve	Ve	Ve
Peak Flow	Peak_Flow	Peak_Flow
Liter Flow	Liter_Flow	Liter_Flow
Ti	Ti	Ti
PEEP	PEEP	PEEP
Pressure Support	Pressure Support	Pressure Support
CPAP	CPAP	CPAP
CMV	CMV	CMV
SIMV	SIMV	SIMV
Flow-by	Flow-by	Flow-by
HFV	HFV	HFV
I:E Ratio	I:E_Ratio	I:E_Ratio
Wave	Wave	Wave
ICD9 Code	ICD9_Code	ICD9_Code
Oxygen Device	Oxygen_Device_1	Oxygen_Device_1
Oxygen Device	Oxygen_Device_2	Oxygen_Device_2
Diagnostic Code	Diagnostic_Code	Diagnostic_Code

C records and NTE segments – general considerations

Parameter error handling in comment records/ note segments

Currently errors associated with a patient, QC or calibration result are included as a ^ separated list of error codes in 1 C record, following the O record or R record, for ASTM transmissions.

When the Audit trail is enabled, each error will have it's own C record/Note segment.

In order to be backward compatible, however, it must still be possible to generate the old format. To do this, the Audit Trail function must be disabled.

In order for the RADIANCE system to be backward compatible, a version of the astm ini file, astm7xx.ini, is installed which can be used to format messages according to the old format.

To use this file the current astm.ini file must be renamed (or deleted) and the file astm7xx.ini must be renamed to astm.ini. The HISLISPlugin must be restarted, via the NTSA, after this change.

Ordering of comment records/ note segments

The comment record/note segment in HIS/LIS transmissions is used to both transmit a log of changes to a service and to document eventual errors/flags associated with parameter values.

When a parameter record/segment has both 1 or more errors and 1 or more changes associated with it then **Error** records/segments **come first** followed by **Change** records/segments.

e.g.

```
R|24|^ ^ ^tO2 ^C|?16.6|Vol%|C|
C|1||210 ^error text 1|
C|2||334 ^error text 2|
C|3||112 ^error text 3|
C|4||CHANGE ^23:05 2001-10-03 (Bill) tO2: 13.2% -> 16.6%|
```

Log of changes to patient/ sample demographic data

If a change has been made to a Patient Result which is not reflected in one of the R records or OBX segments included in the service transmission, (e.g. Patient or Sample Demographics which are located in the P/O record or PID/OBR segments) then the change is documented as a general comment (C record or NTE segment preceding the first R record or OBX segment).

e.g.

```
P|1||112233||Peter ^Hansen||M|||||25 ^years||1.82 ^m|69.0 ^kg|
|
O|1||Sample # ^3|^ ^ ^syringe
|||20011002113012|||||Arterial ^Brachial, left|Dr. McCoy|C|
C|1||CHANGE ^23:14 2001-10-03 (Bill) PatientWeight: 75.0kg ->
69.0kg|
```

Connection failures – general considerations

Handling of missing transmissions during HIS/LIS connection failures

This section specifies what to do if a Patient Result has been received by the RADIANCE system and "edited" or "retransmitted" before there has been a chance to send the Patient Result onwards to the HIS/LIS system for the first time.

This can happen when the HIS/LIS connection has been down for a period of time and one then chooses to send both the "initial" Patient Result and all subsequent "edited" and/or "retransmitted" Patient Results when the connection is re-established.

However, because each transmitted Patient Result contains a complete log of changes, only the latest version of the Patient result is sent and this is marked as "initial" (status Final) in the O record or OBR segment. This enables the HIS/LIS system to handle the service as a first time delivery.

e.g.

```
H|\^&|||ABL735^Central Lab.|||||1|20011003231512
P|1||112233||Peter^Hansen||M|||||25^years||1.82^m|69.0^kg|||||
|
O|1||Sample #^3|^ ^ ^syringe
|||20011002113012|||||Arterial^Brachial, left|Dr. McCoy|||||F|
C|1||CHANGE^23:14 2001-10-03 (Bill) PatientWeight: 75.0kg ->
69.0kg|I
C|2||CHANGE^23:14 2001-10-03 (Bill) pCO2: 111mmHg -> <Not
included>|I
R|1|^ ^ ^Cl^-^M|99|mmol/L||N||R||Bill|20011002114000
R|2|^ ^ ^pH^M|7.402|||N||R|||
R|3|^ ^ ^pO2^M|?111|mmHg||N||R|||
C|1||210|I
R|5|^ ^ ^Na+^M|134|mmol/L||N||R|||
R|6|^ ^ ^Glu^M|5.0|mmol/L||N||R|||
R|7|^ ^ ^Lac^M|1.2|mmol/L||N||R|||
R|8|^ ^ ^Ca++^M|0.54|mmol/L||N||R|||
R|9|^ ^ ^K+^M|3.7|mmol/L||N||R|||
R|10|^ ^ ^tHb^M|10.9|g/dL||N||C|||
C|1||CHANGE^23:04 2001-10-03 (Bill) tHb: 12.8 -> 10.9|I
R|11|^ ^ ^sO2^M|97.5|%||N||R|||
...
L|1|N
```


Audit trail – general considerations

Types of transmissions

The Audit trail of changes to Patient Results are reflected in the HIS/LIS output sent by the RADIANCE system, whether it be HL7 or ASTM. Audit trail is sent via C records for ASTM and via NTE segments for HL7.

NOTICE:

The analyzer does not send Audit trail.

Transmitted Patient Results can be in one of 4 states.

State - Patient Result is considered....	
Initial	When it is received for the first time from the analyzer and does not contain any entries in the change log.
Initial with change log	If it has been received and edited or retransmitted to the RADIANCE system before the service has been sent to the HIS/LIS for the first time. This will typically be the case if the HIS/LIS connection has been down for a period of time. In this case when the connection is recovered the service should be sent as if it had the state "initial" but the entire change log should be included.
Retransmitted	When it has been retransmitted from the analyzer, and it contains changes with respect to the original transmission. These changes will be reflected in the change log in RIME.
Edited	When it has been changed by one of the RADIANCE system applications.

How to send

The "initial" Patient Result should be sent to the HIS/LIS system with the Report type (ASTM) or Result status (HL7) of transmission set to Final status. All subsequent transmissions are considered to be of type "edited" or "retransmitted" and should be sent with status set to Corrected. In addition parameters included in the service will be marked with a status indicating whether they have been changed or not. Refer to the sections ASTM and HL7 below for the specific settings.

When a record, or segment in the transmission, refers to a parameter that has been changed then the change is documented via a series of comment records, or note segments, immediately following the parameter record or segment.

Change log - total and not incremental Each transmission to the HIS/LIS will contain a **COMPLETE** log of changes to the Patient Result since it's "initial" state. This means that the Patient Result will grow in size each time, and that the last transmission will contain all historical information required by the HIS/LIS system to satisfy regulatory requirements. This also implies that not all transmissions are required by the HIS/LIS system to assemble a complete change log for the Patient Result. The last Patient Result can always be manually retransmitted to get a complete log thereby eliminating concern over lost data during retransmission.

Audit trail – ASTM

Marking a patient result as a retransmission

The "Report Type" field (26) in the O record is used to flag a retransmission of a patient result. It may have one of the following values:

F : final result (used for initial transmissions)

C : correction of previously transmitted result (used for all subsequent retransmissions of the same result .

Types of changes marked in retransmission

A change can be an
update
addition
deletion

of a parameter value.

A change in the value, or addition of a new value in a result, is reflected in the reporting of the value. The affected value is marked as changed via the R record for the affected parameter. This is done via the "Result Status" field (9) in the R record. It may have one of the following values:

F: final result (used for initial transmission)

C: Corrected (used for a parameter which has been changed in a retransmission)

R: Retransmission (used for a parameter in a retransmission which has not been changed)

In addition, a series of 1 or more C (comment) records follow the R record for each parameter that has changed. These document the change(s), including the time of change and the operator who has made the change(s). A comment record appears for each change, thereby tracking all changes made, and allowing both the initial result and any intermediate results to be reproduced. The change is specified as free text, and is included as the 2nd component of the "Comment Text" field (4).

For a value which has been deleted, the change is documented by a general comment to the result. These are included as C records and immediately follow the O record but precede the first R record.

For an example transmission refer to *Chapter 5*.

Audit trail – HL7

Marking a result as a retransmission

The "Result status" field (25) in the OBR segment is used to flag a retransmission of a patient result. It may have one of the following values:

F : final result (used for initial transmissions)

C : correction of previously transmitted result (used for all subsequent retransmissions of the same result .

Types of changes marked in retransmission

A change can be an

update

addition

deletion

of a parameter value.

A change in the value or addition of a new value in a result is reflected in the reporting of the value. The affected value is marked as changed via the OBX segment for the affected parameter. This is done via the "Observation Result Status" field (11) in the OBX segment. It may have one of the following values:

F: final result (used for initial transmission)

C: Corrected (used for a parameter which has been changed in a retransmission)

In addition a series of 1 or more NTE (comment) segments follow the OBX segment for each parameter that has changed. These document the change(s), including the time of change and the operator who has made the change(s). A comment segment appears for each change, thereby tracking all changes made, and reproducing both the initial result and any intermediate results. The change is specified as free text, and is included as the 2nd component of the "Comment" field (3).

For a value, which has been deleted, the change is documented by a general comment to the result. These are included as NTE segments and immediately follow the OBR segment but precede the first OBX segment.

For an example of HL7 transmission refer to *HL7 990-493 manual*.

Date of issue

Radiometer representative:

Manufacturer:

RADIOMETER 

If you have any questions
or need assistance,
please contact your local
Radiometer representative.



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